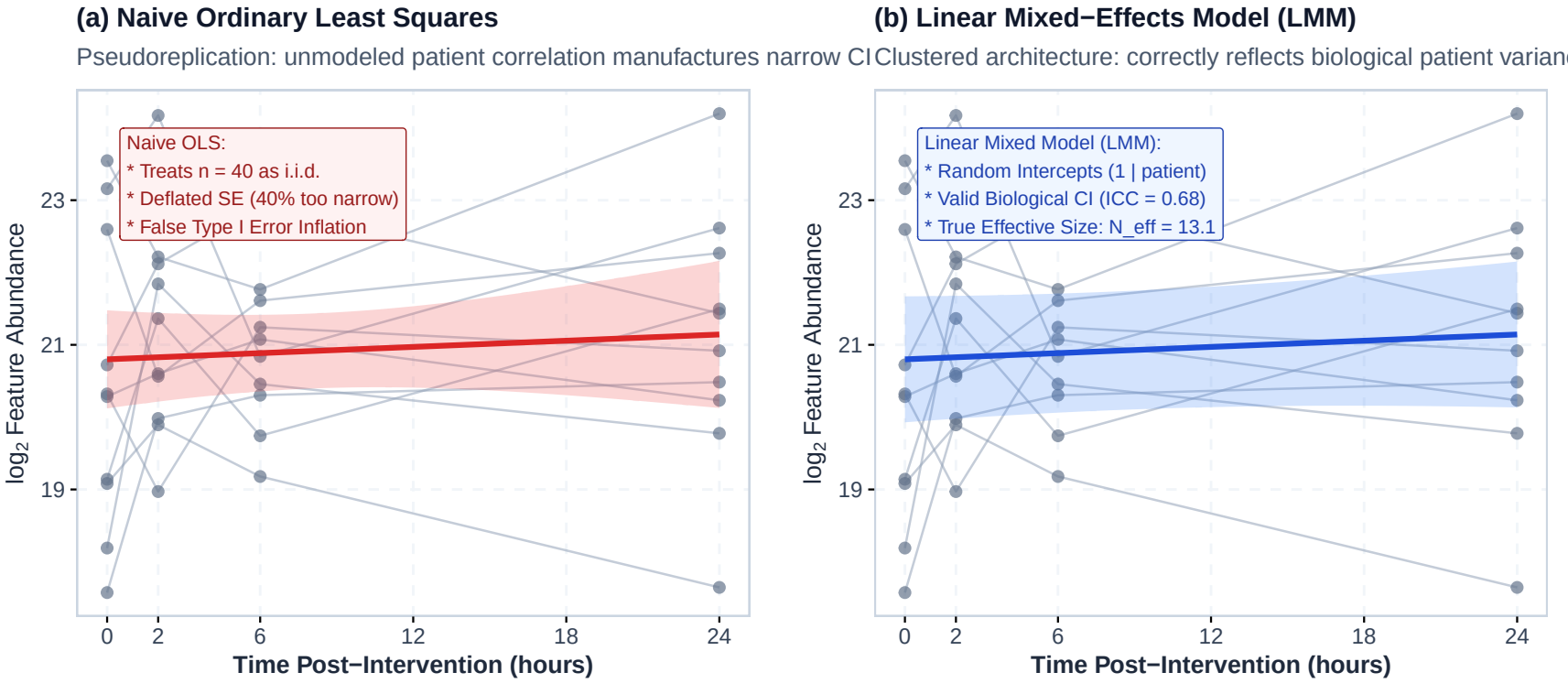
(a) Naive Ordinary Least Squares
Pseudoreplication: unmodeled patient correlation manufactures narrow CI
Naive OLS:
* Treats n = 40 as i.i.d.
* Deflated SE (40% too narrow)
* False Type I Error Inflation
log2 Feature Abundance
23
21
19
0
2
6
12
18
24
Time Post−Intervention (hours)
(b) Linear Mixed−Effects Model (LMM)
Clustered architecture: correctly reflects biological patient variance
Linear Mixed Model (LMM):
* Random Intercepts (1 | patient)
* Valid Biological CI (ICC = 0.68)
* True Effective Size: N_eff = 13.1
log2 Feature Abundance
23
21
19
0
2
6
12
18
24
Time Post−Intervention (hours)